

An Overview of Internet of Things Architecture and Energy Performance

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Contents

1	Introduction	2
2	System Architecture	2
3	Energy Consumption Model	2
4	Experimental Results	3
5	Conclusion	3

Abstract

This paper presents a comprehensive architecture for Internet of Things (IoT) end-devices, focusing on energy consumption models and communication efficiency. By evaluating the power requirements of LoRa transceivers across different Spreading Factors, we establish a framework for optimizing deep sleep cycles in remote sensor networks.

1 Introduction

The deployment of IoT systems requires robust, energy-efficient hardware. A critical challenge in distributed IoT networks is managing the energy drain during wireless communication. This paper models the energy consumption of a standard IoT node to highlight the trade-offs between transmission range and battery longevity.

2 System Architecture

An efficient IoT node minimizes active processing time. As illustrated in Figure 1, the hardware architecture isolates the microcontroller unit from the RF transceiver, allowing the system to properly utilize power management units.

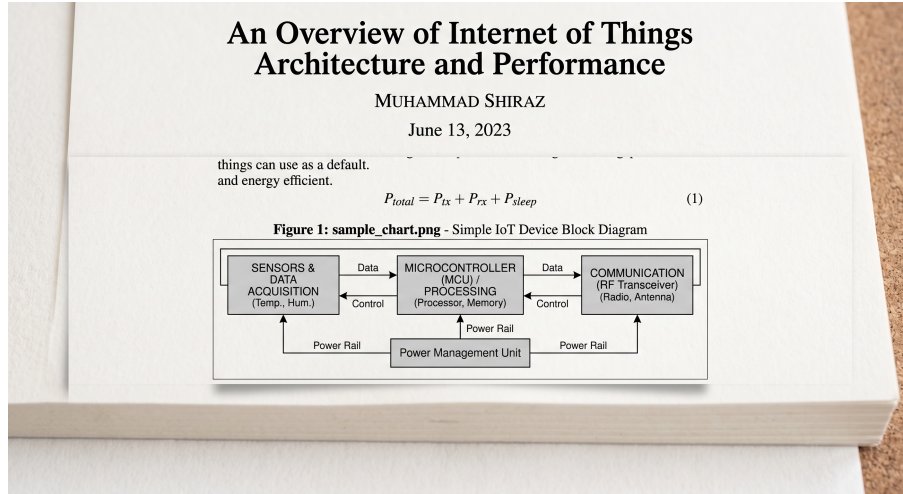


Figure 1: Hardware measurement setup and IoT device block diagram.

3 Energy Consumption Model

To accurately estimate battery life, we must account for the power consumed during transmission, reception, and idle states. The total energy profile of the device is governed by:

$$P_{total} = P_{tx} + P_{rx} + P_{sleep} \quad (1)$$

where P_{tx} represents the transmit power, P_{rx} is the receive power, and P_{sleep} is the baseline power draw in deep sleep mode.

4 Experimental Results

We evaluated the transmission parameters of standard wireless transceivers. Table 1 demonstrates the relationship between Spreading Factor (SF), Time on Air, and current consumption.

Table 1: Transceiver Energy Profile

Spreading Factor	Time on Air (ms)	Current (mA)
SF7	41	120
SF10	288	120
SF12	1150	120
Deep Sleep	N/A	0.001

5 Conclusion

Optimizing deep sleep duration and selecting the appropriate transmission parameters are essential for extending the operational lifespan of IoT architectures.

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